

Supplementary Material

Viewpoint Invariance Is a Theorem, Not a Fit

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This supplement carries the depth condensed out of the main paper: the proof of the invariance theorem (Sec. S1), the full invariant-set specification (Sec. S2), the eight-gate numerical certificate (Sec. S3), the complete precision budget (Sec. S4), the nondeterminism seed distribution (Sec. S5), and the protocol audit and continuous-time control (Sec. S6). All numbers are reproduced from the same banked artifacts as the main paper; equation and proposition references (e.g. Prop. 1) point to the main text.

S1. Proof of the Invariance Theorem

Proposition 1 ($SO(3)$ -invariance of the read-out; = Prop. 1). *Let the encoder intertwine with the Wigner-D action, so under $g \in SO(3)$ each latent transforms as $s_j \mapsto s_j$ (type-0) and $v_j \mapsto D_1(g)v_j$ (type-1), with $D_1(g) \in SO(3)$ orthogonal and $\det D_1(g) = +1$. Then the projection Π (main Eq. 1–2) is invariant under $SO(3)$ for arbitrary encoder weights, and the predicted score is invariant to any rigid camera motion.*

Proof. Feeding root-relative coordinates makes every input a difference $\tilde{x}_i - \tilde{x}_j$, so a camera translation cancels; it remains to treat rotations. Consider each generator of Eq. 1 under $v \mapsto D_1(g)v$ with $R := D_1(g)$ orthogonal: (i) $\tanh s_j$ is a function of a type-0 scalar, hence unchanged. (ii) $\|v_j^c\| = \|Rv_j^c\|$ since $R^\top R = I$; so $\log(1 + \|v\|)$ is invariant. (iii) $\langle \widehat{Rv}, \widehat{Rv'} \rangle = \langle Rv/\|v\|, Rv'/\|v'\| \rangle = \langle \hat{v}, \hat{v'} \rangle$, again because R preserves inner products. (iv) $\det[\widehat{Rv^a}, \widehat{Rv^b}, \widehat{Rv^c}] = \det(R) \det[\hat{v}^a, \hat{v}^b, \hat{v}^c] = \det[\hat{v}^a, \hat{v}^b, \hat{v}^c]$, using $\det(R) = +1$; the triple products are invariant under *proper* rotations (and flip sign under a reflection, $\det = -1$, which is exactly why they narrow the guaranteed group from $O(3)$ to $SO(3)$). The bone lengths $d_{ij} = \|\tilde{x}_i - \tilde{x}_j\|$ and the anatomical signed volumes χ_τ (normalised determinants of joint triads) are invariant by the same orthogonality and $\det(R) = +1$ arguments. Finally mean_j and max_j are applied identically to invariant per-joint quantities, hence themselves invariant. Every entry of Π is therefore fixed by

Table S1. Structural specification of the invariant cut Π .

Generator	Irrep	Count	Dim
<i>Parity-even</i> — generating set complete for $O(3)$			
scalars $\tanh s$	$0e$	n_0	32
vector norms $\log(1 + \ v\)$	$0e$	n_1	8
cosines $\langle \hat{v}, \hat{v'} \rangle$	$0e$	$\binom{n_1}{2}$	28
<i>pooled</i> mean $\oplus$ max		2×68	136
bone lengths d_{ij}	$0e$	per bone	24
even subtotal			160
<i>Parity-odd</i> — $SO(3)$ chiral completion			
triple products $\det[\hat{v}, \hat{v'}, \hat{v'']}$	$0o$	$\binom{n_1}{3}$	56
<i>pooled</i> mean $\oplus$ max		2×56	112
anatomical volumes χ_τ	$0o$	per frame	11
odd subtotal			123
full $SO(3)$ cut			283

g , for any weights; the head is a deterministic function of Π , so the score is invariant. $\square$

S2. Full Invariant-Set Specification

Table S1 states both generating sets explicitly at the deployed widths ($n_0=32$, $n_1=8$; $J=25$ joints, 24 bones, $|\mathcal{T}|=11$ anatomical frames). Both parity classes share one mean $\oplus$ max joint pool; bone lengths and signed volumes are concatenated index-preserved (unpooled). The parity-even subtotal (160) is exactly the projection whose precision budget is audited in Sec. S4; the parity-odd completion adds 123 dimensions.

S3. Numerical Certificate (E1–E8)

We certify the *deployed* model. E1–E5 establish that the claimed invariance is present; E6–E8 establish that it is the one we actually have. The three conditions of E3–E4 — violation at roundoff, *flat* in angle, and $r \gg 1$ — are what upgrade “small” to “zero”: magnitude alone would not. Reported figures are from a representative fp64 run; being roundoff-floor quantities they are *regime*-stable (order of

- 057 magnitude, sign of slope, $r \gg 10^5$) but not bit-reproducible.
 058 E1 **Representation orthogonality:** $\max_g \|\rho(g)^\top \rho(g) -$
 059 $I\|_F \leq 10^{-12}$ (fp64).
 060 E2 **Invariance of the motion channels:** δ, σ verified invari-
 061 ant, not assumed.
 062 E3 **Rotation sweep:** score violation $\sim 10^{-16}$ across $\theta \in$
 063 $[0.05, 3.0]$ rad, with log-log slope $|\text{slope}| < 0.5$ (flat).
 064 A modelling violation grows with angle; roundoff does
 065 not.
 066 E4 **Precision scaling:** $r = \bar{\delta}_{\text{fp}32} / \bar{\delta}_{\text{fp}64} = 5.44 \times$
 067 $10^{-8} / 4.43 \times 10^{-16} = 1.23 \times 10^8$, tracking the machine-
 068 epsilon ratio; a genuine violation would be precision-
 069 independent ($r \approx 1$).
 070 E5 **Translation invariance:** on raw world coordinates,
 071 $|s(\mathbf{x} + \mathbf{c}) - s(\mathbf{x})| / |s(\mathbf{x})| = 2.6 \times 10^{-16}$.
 072 E6 **Parity:** under a reflection the parity-even cut gives *ex-*
 073 *actly* 0 (bit-identical on a mirrored body — an unin-
 074 tended $O(3)$ -invariance), whereas the completed cut gives
 075 4.4×10^{-2} (seed 0, `certify_egru.json`). A single
 076 reflection certifies the entire improper coset.
 077 E7 **Laterality:** permuting left/right joint *indices* (a rela-
 078 belling, not a reflection) moves the score by 6.9×10^{-2}
 079 / 5.6×10^{-2} for the two cuts (seed 0); both separate
 080 a left- from a right-limb movement. E6–E7 are model-
 081 dependent modelling quantities, not roundoff-floor ones,
 082 and vary across seeds.
 083 E8 **The fix is free:** read-out invariance under Haar-random
 084 *proper* rotations (up to $\theta = \pi$), with the parity-odd chan-
 085 nels active, is 2.9×10^{-16} — unchanged. E1–E5 are
 086 likewise over Haar-random $SO(3)$, so the certified in-
 087 variance is to the full rotation group, of which a camera
 088 azimuth or the NTU cross-view displacement is one in-
 089 stance (5.8×10^{-9} , fp64).

090 S4. Complete Precision Budget

091 The full fp64→int8 budget — including the bf16 and int8-
 092 activations-only rows — is reported in the main paper’s de-
 093 ployment table. The int8 cliff is attributable *solely* to ac-
 094 tivation grid-rounding: weight-only quantization preserves
 095 the theorem (a quantized model is a different equivariant
 096 model), and fp16 is $3.2\times$ tighter than bf16 because equiv-
 097 ariance pays in mantissa bits, which bf16 trades for expo-
 098 nent range.

099 S5. Nondeterminism: Seed Distribution

100 The 0.33 MAD floor quoted in the main paper is the tighter
 101 *fixed-configuration* run-to-run component (init held con-
 102 stant, atomic nondeterminism only). Table S2 gives the
 103 wider *seed-to-seed* distribution (SO(3) EGRU clean MAD
 104 per fold across three initializations): mean across-seed std
 105 0.48, worst pairwise gap 1.30. Both exceed every model-vs-

Table S2. Per-(seed,fold) clean MAD, SO(3) EGRU. No retraining beyond the three seeds.

fold	seed 0	seed 1	seed 2	std	gap
0	7.120	6.778	6.737	0.210	0.383
1	6.680	6.124	5.804	0.443	0.876
2	6.976	7.345	6.623	0.361	0.722
3	6.762	7.363	6.093	0.635	1.270
4	6.469	7.714	6.415	0.735	1.299
mean across-seed std				0.48	

model clean gap in the main accuracy table, so those gaps
 are ties. cuDNN-deterministic modes remove the cuDNN-
 GRU atomic component but not the initialization variance;
 the structural degradation columns are weight-independent
 (Prop. 1 holds for arbitrary weights), so the flat viewpoint
 curve stayed exactly flat across every seed while absolute
 MAD moved.

S6. Protocol Audit and Continuous-Time Control

Protocol audit. The single-exercise audit — selection bias
 $+1.2 \pm 0.4$ MAD (18.6% over three seeds), paired subject-
 cluster bootstrap $\Delta = -0.44$ (per-seed $[-1.07, +0.20]$,
 clearing zero in only one of three seeds) — is reported per-
 seed in the main paper. It is *not* subject leakage (each sub-
 ject contributes one recording on that slice, so a random
 split is already subject-disjoint); the slice simply carries no
 robust subject-level signal, so we pool the five exercises
 and gate every experiment on a per-exercise mean-predictor
 floor.

Continuous-time control. We implement the Neural
 CDE variant and certify it to the same E1–E8 standard; it
 satisfies an exact equivariance certificate and *fails* its own
 measured floor (8.43 vs. 8.25 MAD), the cleanest demon-
 stration that a structural guarantee is worthless without mea-
 sured signal. Analysing the adaptive solver, a Dormand-
 Prince controller accepts/rejects steps on a per-component
 scaled-RMS norm that is *not* group-invariant even when
 the vector field is exactly equivariant, so rotated and unro-
 tated integrations receive different step grids and de-align;
 a per-irrep-normalised norm removes this, driving grid di-
 vergence to identically zero.

Irregular-sampling null (derived). A model-free spec-
 tral census finds 97.7% of KIMORE’s *positional* energy be-
 low the fixed-grid resampling corner (though 70% of *ve-*
locity energy lies above it), so resampling preserves pose
 geometry while low-passing the derivative band and acts
 as a denoiser on healthy-form exercises, where the dis-
 criminative signal is positional — the null was predictable
 before the experiment, though the boundary would move

for tremor-dominated tasks. The mechanism is nonetheless 145
real: a physiologically calibrated 5 Hz tremor is recovered 146
from the true irregular stamps at $r=1.000$ under drop rates 147
up to 70%, while a resampled estimator decays to 0.785. 148